

- 19 B** Since oil drop is stationary, weight of oil drop is equal to electric force.

$$mg = q \frac{V}{d} = 3.2 \times 10^{-19} \times \frac{5000}{0.016} = 1.0 \times 10^{-13} \text{ N}$$

$$mg - q \frac{V'}{d} = ma$$

$$1.0 \times 10^{-13} - \left(3.2 \times 10^{-19} \times \frac{4000}{0.016} \right) = \frac{1.0 \times 10^{-13}}{9.81} \times a$$

$$a = 1.96 \text{ m s}^{-2}$$

OR

$$mg = q \frac{V}{d} = q \frac{5000}{d}$$

$$F_E' = q \frac{4000}{d} = \frac{4}{5} mg = 0.8mg$$

$$mg - 0.8mg = ma$$

$$a = 1.96 \text{ m s}^{-2}$$

- 20 B** Since current is flowing through the specimen, current must be the same in each section